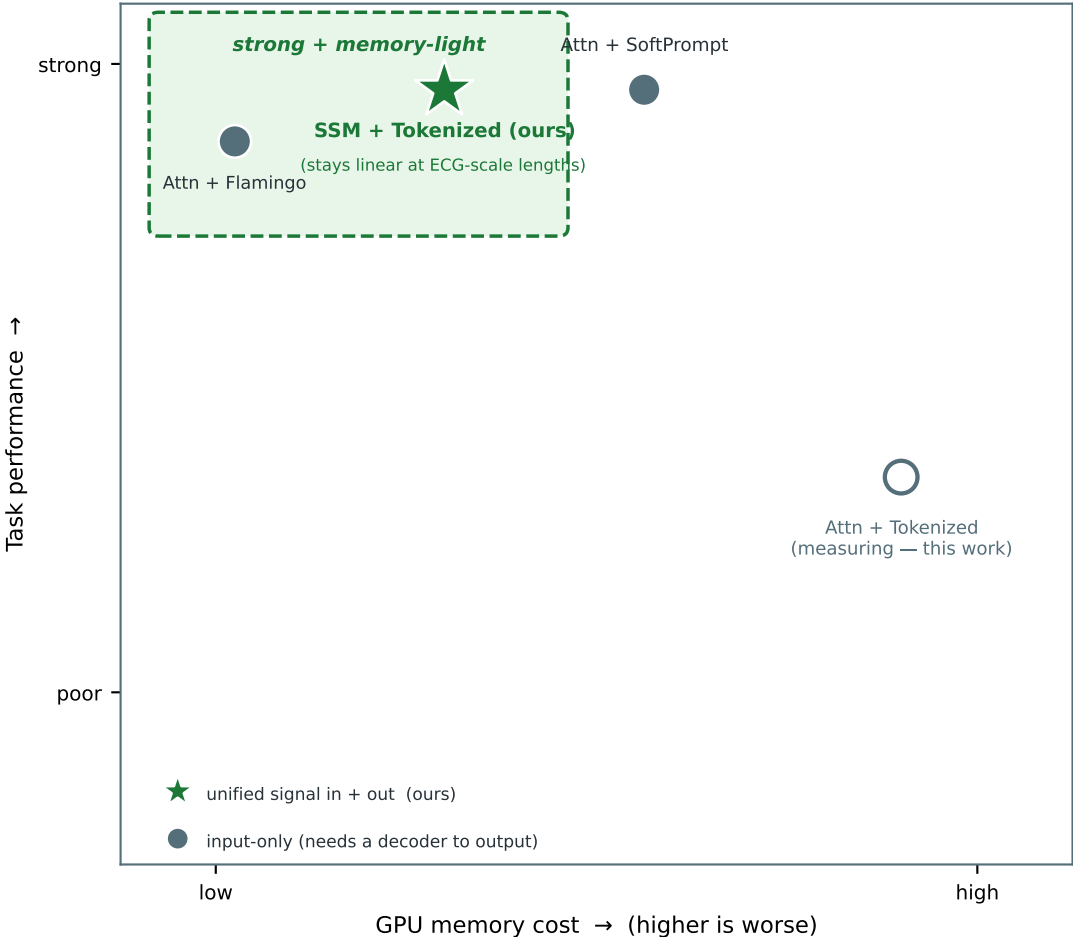


A unified time-series LLM: tokenized signal on an SSM backbone

unified signal in + out needs tokenization — quadratic compute for attention, natural for a linear-time SSM

Where each approach sits



Capabilities at a glance

Approach	Task performance	Memory	Unified in + out?
Attn + Flamingo	strong	low / flat (cross-attn)	no (input-only)
Attn + SoftPrompt	strong	grows w/ len (patch tokens)	no (input-only)
Attn + Tokenized	pending	high at length ($O(L^2)$ compute)	yes
SSM + Tokenized (ours)	strong	low ($O(L)$ / $O(1)$ state)	yes

Memory at typical task lengths; the separation appears at long signals (ECG-scale), where encoder rows stay flat only by downsampling and attention+tokens pays quadratic compute.